Python and Bash Assignment

1. Grade Checker:-

1. **Start**
2. **Input Marks**: Prompt the user to enter their marks.
3. **Check Conditions Using If-Else**:
   - If marks are greater than or equal to 90, assign grade “A”.
   - Else if marks are greater than or equal to 80, assign grade “B”.
   - Else if marks are greater than or equal to 70, assign grade “C”.
   - Else if marks are greater than or equal to 60, assign grade “D”.
   - Else, assign grade “F”.

Code:

```
marks = int(input("Enter your marks: "))

if marks >= 90:
 print("Grade: A")
elif marks >= 80:
 print("Grade: B")
elif marks >= 70:
 print("Grade: C")
elif marks >= 60:
 print("Grade: D")
else:
 print("Grade: F")
```

Conclusion:

The program successfully calculates grade based on marks.

.vscode > Project.py > ...

```python
marks = int(input("Enter your marks: "))

if marks >= 90:
    print("Grade: A")
elif marks >= 80:
    print("Grade: B")
elif marks >= 70:
    print("Grade: C")
elif marks >= 60:
    print("Grade: D")
else:
    print("Grade: F")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shakt\OneDrive\Desktop\UNIVERSE\PYTHON> python -u "c:\Users\shakt\OneDr
Enter your marks: 10
Grade: F
PS C:\Users\shakt\OneDrive\Desktop\UNIVERSE\PYTHON>
```

2. Student Grades (Dictionary)

Aim:

To store and manage student grades using dictionary.

Algorithm:

1. Start
2. Create dictionary
3. Add/Update student
4. Display records
5. Stop

Conclusion:

Dictionary operations performed successfully.

Code:

```
students = {}

while True:
print("1.Add 2.Update 3.Show 4.Exit")
choice = input("Enter choice: ")

if choice == "1":
name = input("Name: ")
grade = input("Grade: ")
students[name] = grade

elif choice == "2":
name = input("Enter name: ")
if name in students:
students[name] = input("New grade: ")

elif choice == "3":
for n, g in students.items():
print(n, g)

elif choice == "4":
break
```

```python
students = {}

while True:
    print("\n1. Add Student")
    print("2. Update Student")
    print("3. Show All")
    print("4. Exit")

    choice = input("Enter choice: ")

    if choice == "1":
        name = input("Enter student name: ")
        grade = input("Enter grade: ")
        students[name] = grade

    elif choice == "2":
        name = input("Enter student name to update: ")
        if name in students:
            students[name] = input("Enter new grade: ")
        else:
            print("Student not found")

    elif choice == "3":
        for name, grade in students.items():
            print(name, ":", grade)

    elif choice == "4":
        break

    else:
        print("Invalid choice")
```

OBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
C:\Users\shakt\OneDrive\Desktop\UNIVERSE\PYTHON> python -u "c:\Users\shakt\OneDrive\Desktop\UNIVERSE\PYTHON\.vscode\Student Grade.py
Add Student
Update Student
Show All
Exit
ter choice: 1
ter student name: Shakti
ter grade: A

Add Student
Update Student
Show All
Exit
ter choice:
```

3. **Write to File**

Aim:

To write data into a file.

Code:

```
file = open("data.txt","w")
file.write("Hello World")
file.close()
```

Conclusion:

Data written successfully.

4. Read from File

Aim:

To read data from a file.

Code:

```
file = open("data.txt","r")
print(file.read())
file.close()
```

Conclusion:

File read successfully.